



Machine Vision

A Focus On IMAGE PROCESSING With Neural Networks



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Some time ago there were some very interesting product releases from the likes of Google and Microsoft. While Google announced a new assistant in Google Allo, which is quite friendly, Microsoft announced a puck-like device for artists working on Surface Pro devices. Google's chat assistant on my phone read my personal messages and gave suggestions in reply.

Is this just a matter of word association, or an invasion of privacy? However, my intrigue began when a friend sent an image of his presence at a club. Allo showed some automatic reply options including, "Have a good time!" To me, it seemed like a fun and convenient way to reduce the hassle of typing out a full message.

Analysing images is a serious task

Photography is one area that employs image analytics heavily. Dr Joseph Reger, chief technology officer, Fujitsu EMEA, explains, "Modern cameras identify faces to make focusing easier, or these can identify smiling faces or faces of pre-selected people." Image analysis starts by first identifying edges and moves on to full shapes. However, making changes in certain key areas in the images has successfully fooled such systems. More on that later, first let us look

at how a computer recognises and processes images.

Image analysis ranges from finding basic shapes, detecting edges, removing noise and counting objects, to finding anomalies in the routine. Security and surveillance are incomplete without an intelligent system that can detect pedestrians and vehicles, and make a difference. So how does image analytics work? How was Google's new assistant able to analyse images and suggest possible replies as soon as I received the image?

Uncanny Vision, a startup focused on image analysis, helps clarify how it happens. Their system identifies objects in images using an analytical system powered by deep learning and smart algorithms. The deep learning framework, Caffe, powers this system. The framework managed by Berkeley Vision and Learning Center (BVLC) is capable of processing more than 60 million images in a single day using just one NVIDIA K40 GPU. On an average, this amounts to about 695 images per second, whereas we only need 17 to make a series of images into a video, with HD video going up to 60. Basically, this system can analyse about 11 HD videos at once.

Deep learning is affecting image analytics. Deep learning is a branch of machine learning based on a set of algorithms that attempt to model high-level abstractions in data by using a deep graph with multiple processing layers, linear and non-linear. It is a representational learning approach ideally suited for image analysis challenges ranging from security and daily-purpose applications to medical problems.

Object detection is considered to be the most basic application of computer vision. Enhancements on top of object detection result in other developments including anomaly mapping and error detection in real-time imaging. However, deep learning

Artificial neural networks process information the way biological nervous systems like the brain do, for optimal decision making. Certain characteristics of their architecture and the way these process information makes these superior to conventional techniques on a certain class of optimisation problems.

Optimisation techniques are called into play every day for industrial planning, allocation, scheduling and decision-making, among others. So how do you optimise the network?

- Select a network architecture, that is, number of hidden layers, number of neurons in each layer and activation function.
- Initialise weights randomly (weights being the mathematical formula).
- Use forward propagation to determine the output node.
- Find the error of the model using known labels.
- Back-propagate the error into the network and determine the error for each node.
- Update the weights to minimise gradient.